Exploratory Data Analysis Report

Iris Dataset

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1. Distribution of Numerical Features (Histograms)

Histograms were plotted for the numerical features:

- Sepal Length
- Sepal Width
- Petal Length
- Petal Width

Insights

- **Sepal Length** shows a near-normal distribution with slight right skew.
- **Sepal Width** is moderately normally distributed but slightly left-skewed.
- **Petal Length** shows clear separation between species, indicating strong predictive power.
- **Petal Width** also shows distinct clusters, making it highly informative.

📌 *Petal-based features display clearer class separation than sepal-based features.*

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2. Analysis of Categorical Feature (Count Plot)

A count plot was used to visualize the categorical feature:

- **Species** (Setosa, Versicolor, Virginica)

Insights

- Each species has an **equal number of samples (50 each)**.
- No class imbalance is present.
- Balanced classes make the dataset ideal for classification models.

📌 *Balanced target variable improves model fairness and evaluation.*

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3. Outlier Detection Using Box Plots

Box plots were created for all numerical features.

Insights

- **Sepal Width** contains a few mild outliers.
- **Petal Length** and **Petal Width** show minimal to no extreme outliers.
- Outliers are not severe enough to require removal.

📌 *The dataset is clean and robust, requiring minimal preprocessing.*

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4. Correlation Heatmap

A correlation heatmap was plotted to examine relationships between numerical features.

Insights

- **Petal Length & Petal Width** have a **strong positive correlation**.
- **Sepal Length** shows moderate correlation with petal features.
- **Sepal Width** has weak correlation with other variables.

📌 *Highly correlated features indicate shared information useful for prediction.*

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5. Visualization-Based Insights Summary

Visualization	Key Insight
Histograms	Petal features show clear separation
Count Plot	Dataset is perfectly balanced
Box Plot	Minimal outliers present
Heatmap	Petal features are highly correlated

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6. Important Features for Prediction

Based on visual analysis:

- ✅ **Petal Length**
- ✅ **Petal Width**

These features:

- Clearly separate species
- Show strong correlation
- Have minimal overlap across classes

⚠ **Sepal Width** is least informative due to overlap and weak correlation.

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7. Final Summary (Bullet Points)

- Iris dataset is **clean, balanced, and well-structured**.
- No missing values or serious outliers detected.
- Petal features are the **most important predictors**.
- Sepal features provide supplementary information.
- Dataset is ideal for:
    - Logistic Regression
    - KNN
    - Decision Trees
    - Random Forest
- Minimal preprocessing required before model training.